



Diet Diversity Analysis of Adults in US

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Background

Diet Diversity : variety of food consumed across food groups

Used to measure diet quality and nutritional adequacy

High diet diversity has been associated with:

- Improving nutrition status
- Better gut health
- Lowering health risk
- Preventing diseases

Research question:

Which demographic factors influence diet diversity the most?

How does each affect diet diversity?



Data and Methods

National Health and Nutrition
Examination Survey (NHANES)
2021-2023

Datasets

- Demographics
- Dietary Interview (Two Days Dietary Log)
- Food Security Questionnaire

Final Dataset:

- 3798 individuals

Explanatory Variables

- Age
- Gender
- Education
- Income to poverty ratio
- Race
- Country of Birth
- Household size
- Marital status
- Food security

Response Variable

Diet Diversity Score: Average number of food groups between day 1 and day 2
(Counted by unique 1st digit of food group code)

Statistical Analysis- Linear Regression

Linear Model:

DDS ~Age + Gender +
Education + Income +
Race + Country of Birth
+ Household size +
Marital status + Food
security

Residuals:

Min	1Q	Median	3Q	Max
-3.8115	-0.7375	0.0292	0.7971	3.3352

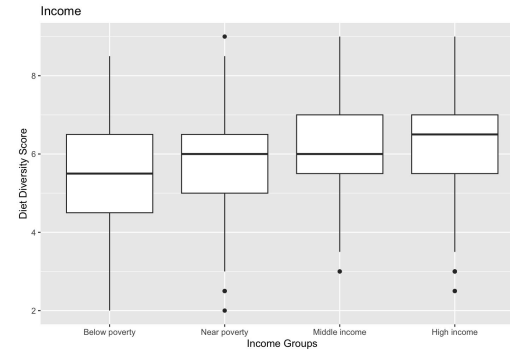
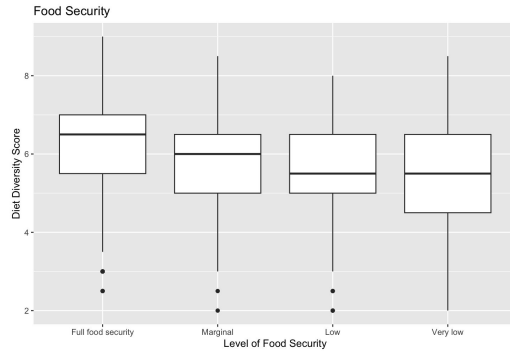
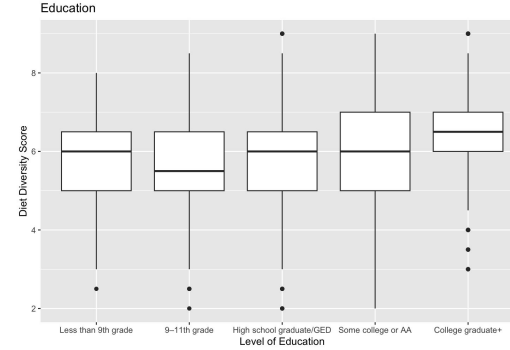
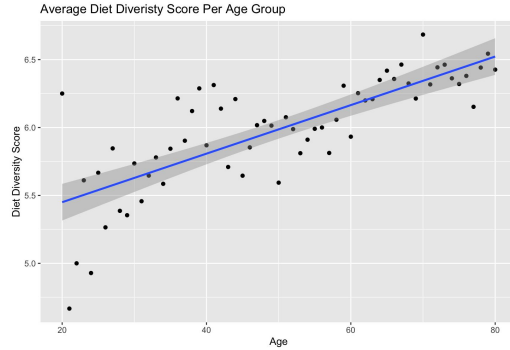
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.77488	0.14966	38.587	< 2e-16 ***
DMDEDUC29-11th grade	0.20351	0.14102	1.443	0.149090
DMDEDUC2High school graduate/GED	0.31178	0.12938	2.410	0.016017 *
DMDEDUC2Some college or AA	0.41875	0.12751	3.284	0.001035 **
DMDEDUC2College graduate+	0.68977	0.13092	5.269	1.47e-07 ***
RIAGENDRFemale	0.10916	0.04247	2.570	0.010208 *
DMDBORN4Non-US	0.26102	0.07062	3.696	0.000223 ***
DMDMARTZWidowed/Divorced/Separated	-0.17722	0.05481	-3.233	0.001236 **
DMDMARTZNever married	-0.12496	0.06443	-1.940	0.052523 .
RIDRETH3Other Hispanic	-0.10579	0.11232	-0.942	0.346361
RIDRETH3Non-Hispanic White	-0.16242	0.10212	-1.590	0.111843
RIDRETH3Non-Hispanic Black	-0.26354	0.11352	-2.322	0.020322 *
RIDRETH3Non-Hispanic Asian	-0.33626	0.13727	-2.450	0.014354 *
RIDRETH3Other / Multi-racial	-0.17033	0.12604	-1.351	0.176670
FSDADMarginal	-0.04541	0.07599	-0.598	0.550149
FSDADLow	-0.21718	0.07934	-2.737	0.006232 **
FSDADVery low	-0.35531	0.08452	-4.204	2.70e-05 ***
RIDAGEYR_s	0.28492	0.02475	11.510	< 2e-16 ***
INDFMPIR_s	0.18408	0.02722	6.762	1.62e-11 ***
DMDHHSIZ_s	0.01283	0.02457	0.522	0.601534

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.135 on 3018 degrees of freedom
Multiple R-squared: 0.1682, Adjusted R-squared: 0.163
F-statistic: 32.12 on 19 and 3018 DF, p-value: < 2.2e-16

Statistical Analysis - Linear Regression



Statistical Analysis - Lasso

Lasso: selects the most important variables by shrinking less important ones to zero, simplifying the model and improving interpretability

- Strong predictors: Education, US born, Food Security, Income
- Household size, Hispanic and White race categories, Never married dropped.

	Estimate
(Intercept)	5.927791487
DMDEDUC29-11th grade	-0.049120854
DMDEDUC2High school graduate/GED	.
DMDEDUC2Some college or AA	0.098939918
DMDEDUC2College graduate+	0.371004853
RIAGENDRFemale	0.088179985
DMDBORN4Non-US	0.230230723
DMDMARTZWidowed/Divorced/Separated	-0.155182413
DMDMARTZNever married	-0.119524898
RIDRETH3Other Hispanic	.
RIDRETH3Non-Hispanic White	.
RIDRETH3Non-Hispanic Black	-0.092495741
RIDRETH3Non-Hispanic Asian	-0.108979327
RIDRETH3Other / Multi-racial	.
FSDADMarginal	-0.016206781
FSDADLow	-0.187903950
FSDADVery low	-0.332141680
RIDAGEYR_s	0.263691288
INDFMPIR_s	0.189595457
DMDHHSIZ_s	0.001737936

Statistical Analysis - Random Forest

Random Forest:

Use multiple decision trees to prevent overfitting and increase accuracy

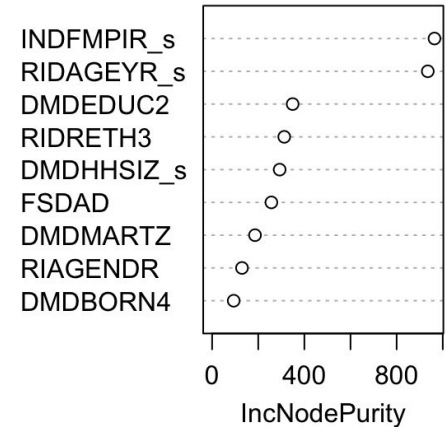
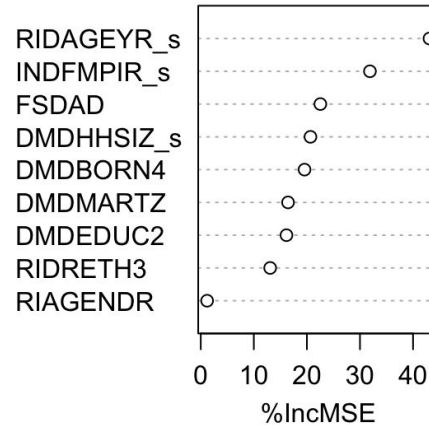
Model:

DDS ~ Age + Gender + Education + Income + Race + Country of Birth +
Household size + Marital status + Food security

$m = 3$

Statistical Analysis - Random Forest

	%IncMSE	IncNodePurity
DMDEDUC2	17.289069	307.95733
RIAGENDR	-1.415988	115.23852
RIDAGEYR	34.471456	832.92726
DMDBORN4	20.060761	86.69975
DMDMARTZ	16.141963	91.51905
DMDHHSIZ	17.655225	263.93372
INDFMPIR	22.914246	891.60100
RIDRETH3	19.935420	263.60383
FSDAD	22.560117	227.45794



- Age and Income most important variables
- Gender doesn't affect diet diversity.

Conclusion

- Higher food security, income, education → More diverse diets
- Age and Income are key factors
- Improve access of diverse foods and nutritional information for groups with lower diet diversity.

Limitations

- A lot of missing data and not representative of the US population
- Inaccurate filling of survey

References

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: With applications in R. Springer.

Centers for Disease Control and Prevention. (n.d.). Dietary analyses tutorials. National Health and Nutrition Examination Survey (NHANES). <https://wwwn.cdc.gov/nchs/nhanes/tutorials/DietaryAnalyses.aspx>

Centers for Disease Control and Prevention. (n.d.). NHANES 2021–2023 datasets and documentation. National Health and Nutrition Examination Survey (NHANES).

<https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2021-2023>

Zhao, W., Zhang, J., Zhao, A., Wang, P., & Zhang, Y. (2020). Dietary diversity scores: An indicator of micronutrient inadequacy instead of obesity for Chinese children. BMC Public Health, 20, 462. <https://doi.org/10.1186/s12889-020-8451-7>

de Oliveira Santos, M., et al. (2024). Dietary diversity and associated socioeconomic factors among adults: A population-based analysis. Nutrients. Advance online publication. <https://doi.org/10.3390/nu16193334>

Hossain, M. B., et al. (2023). Dietary diversity and nutritional adequacy among adults: Evidence from population studies. Nutrients, 15(9), 2154. <https://doi.org/10.3390/nu15092154>